



MIRepNet: A pipeline and pre-trained model for EEG-based motor imagery classification

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ABSTRACT

Brain-computer interfaces (BCIs) enable direct communication between the brain and external devices. Recent EEG foundation models aim to learn generalized representations across diverse BCI paradigms. However, these approaches overlook fundamental paradigm-specific neurophysiological distinctions, limiting their generalization ability. Importantly, in practical BCI deployments, the specific paradigm such as motor imagery (MI) for stroke rehabilitation or assistive robotics, is generally determined prior to data acquisition. This paper proposes MIRepNet (Rep = Representations), the first EEG pre-trained model (PTM) tailored for the MI paradigm. MIRepNet comprises a high-quality EEG preprocessing pipeline incorporating a neurophysiologically-informed channel template, adaptable to EEG headsets with arbitrary electrode configurations. Furthermore, we introduce an effective pre-training strategy that combines self-supervised masked token reconstruction and supervised MI classification, facilitating rapid adaptation and accurate decoding on novel downstream MI tasks with fewer than 30 trials per class. Extensive evaluations across five public MI datasets demonstrated that MIRepNet consistently achieved state-of-the-art performance, significantly outperforming both specialized and generalized EEG models. Our code is available at GitHub¹.

1. Introduction

A brain-computer interface (BCI) establishes a direct communication pathway between the brain and external devices [1]. BCIs have been widely adopted in diverse applications, including brain disease diagnosis, neuro-rehabilitation, and assistive device control [2,3]. Among the various neural recording modalities, Electroencephalogram (EEG) is one of the most widely used non-invasive modalities for BCIs due to its cost-effectiveness, ease of use, and safety [4,5]. However, the practical deployment of EEG-based BCIs faces significant challenges, including substantial individual differences, limited availability of high-quality training data, and discrepancies in electrode configurations across different EEG headsets [6,7]. Consequently, traditional approaches typically require extensive calibration data collection from each user, posing a substantial obstacle to real-world usability.

Recently, generalized EEG foundation models [8–15] have emerged as promising solutions for addressing calibration challenges by learning universal EEG representations from large-scale datasets. Such foundation models aim to facilitate effective adaptation across diverse downstream BCI tasks. However, existing EEG generalized models face several inherent limitations:

1. Existing models typically merge EEG signals from various paradigms, such as motor imagery (MI), steady-state visual evoked potentials (SSVEP), and event-related potentials (ERP), during pre-training. Nevertheless, distinct BCI paradigms inherently involve fundamentally different neurophysiological mechanisms, characterized by paradigm-specific active cortical regions and frequency bands [16]. Combining multiple paradigms into a single pre-training stage contradicts the intrinsic neurophysiological organization of EEG signals, resulting in suboptimal

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¹ <https://github.com/staraink/MIRepNet>

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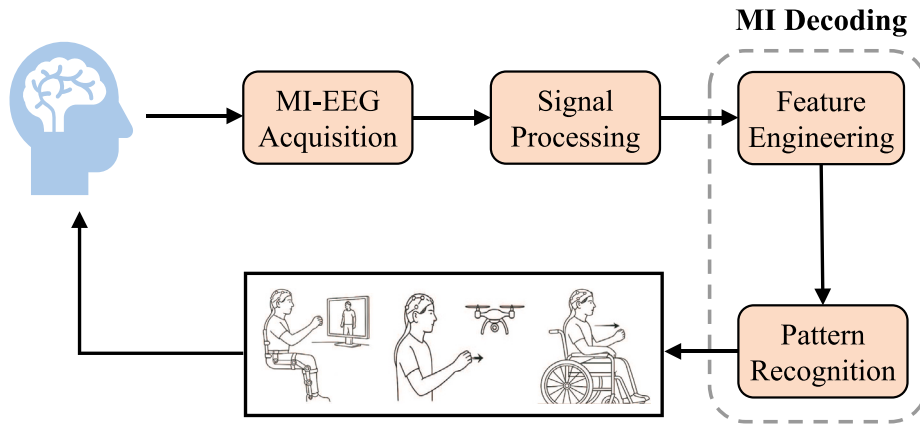


Fig. 1. A closed-loop MI-based BCI system.

- representations and limited downstream generalization performance.
2. In practice, the paradigm associated with a downstream task is typically predetermined before data acquisition. For instance, stroke patients requiring exoskeleton assisted rehabilitation naturally align with an MI-based system [17], whereas epilepsy monitoring necessitates a dedicated epilepsy-specific paradigm and corresponding model [18]. Consequently, when the user profile encompassing patient population and intended task is accessible, selecting a paradigm-specific foundation (pre-trained) model for direct adaptation becomes both feasible and preferable.
 3. More critically, current generalized models require extensive post-training involving other subjects' data from the entire downstream dataset (usually comprising 8–12 subjects) to adapt effectively to a new subject. Compared to a scenario in which only 30% of a single new subject's data is needed for fine-tuning, the adaptation cost in terms of data volume increases dramatically (at least 20×). Such extensive post-training may even underperform conventional subject-specific training on the same data (see Fig. 2(a)). Ideally, a foundation model should adapt directly to any new subject or task without additional large-scale post-training [19].

These insights motivate the development of paradigm-specific foundation (pre-trained) models that support efficient adaptation using limited data from a new user. Such models promise significantly enhanced decoding accuracy and substantially reduced calibration requirements. Among existing BCI paradigms, MI is particularly noteworthy due to its extensive exploration and broad clinical applications, including stroke rehabilitation and assistive technologies such as smart wheelchairs [20].

This paper proposes *MIRepNet*, an EEG pre-trained model explicitly designed for MI (whose pipeline is shown in Fig. 1). *MIRepNet* leverages neurophysiological insights of the MI paradigm to construct a tailored preprocessing and representation learning pipeline, significantly enhancing the generalizability and efficiency of EEG decoding.

The main contributions of this work are:

1. We introduce *MIRepNet*, the first pre-trained model specifically tailored for MI tasks. By capturing MI-specific neurophysiological features, *MIRepNet* effectively learns generalizable representations for MI decoding.
2. We propose a high-quality EEG preprocessing pipeline comprising subject screening, a unified channel template-based spatial alignment, frequency filtering, temporal resampling, and distribution alignment. This approach addresses challenges arising from heterogeneous EEG headset configurations, ensuring data consistency across diverse datasets.

3. We develop an efficient pre-training approach combining masked token reconstruction and supervised MI classification. This strategy enables the model to acquire robust, generalizable temporal-spatial EEG representations.

Extensive experiments on five public MI datasets including 47 downstream subjects demonstrated that *MIRepNet* achieved state-of-the-art decoding accuracy (see Fig. 2(b)). Moreover, *MIRepNet* required significantly fewer calibration trials (fewer than 30 trials per class) and rapidly converged in a few epochs, highlighting its practical utility and effectiveness.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 proposes *MIRepNet*. Section 4 presents the experiment results. Section 5 introduces the discussion. Finally, Section 6 draws conclusions.

2. Related works

This section introduces related works on traditional MI decoding algorithms and EEG generalized models.

2.1. MI decoding algorithms

Convolutional neural networks (CNNs) have achieved significant success in EEG decoding. EEGNet [21] utilizes compact depthwise separable convolutions to efficiently extract discriminative EEG features. ShallowConvNet and DeepConvNet [22] integrate temporal and spatial convolutions with nonlinear transformations to capture log-band-power features. FBCNet [23] employs filter-bank spectral filtering combined with depthwise spatial convolutions, effectively modeling spectro-spatial-temporal patterns. IFNet [24] explicitly captures cross-frequency interactions via interactive frequency convolution modules, while EEG-Conformer [25] merges local convolutional features with global self-attention mechanisms. However, these specialized methods typically require extensive labeled data from individual subjects or specific acquisition setups, limiting their generalizability and practical deployment.

2.2. EEG generalized models

Recent EEG foundation models seek to achieve robust transferability across diverse downstream tasks. BIOT [9] tokenizes multi-channel biosignals into unified “sentences” and employs Transformer-based cross-dataset pre-training. BENDR [26] utilizes a wav2vec-style self-supervised Transformer for contrastive representation learning directly from large-scale raw EEG signals. CBraMod [10] employs a criss-cross transformer with masked EEG reconstruction. Similarly,

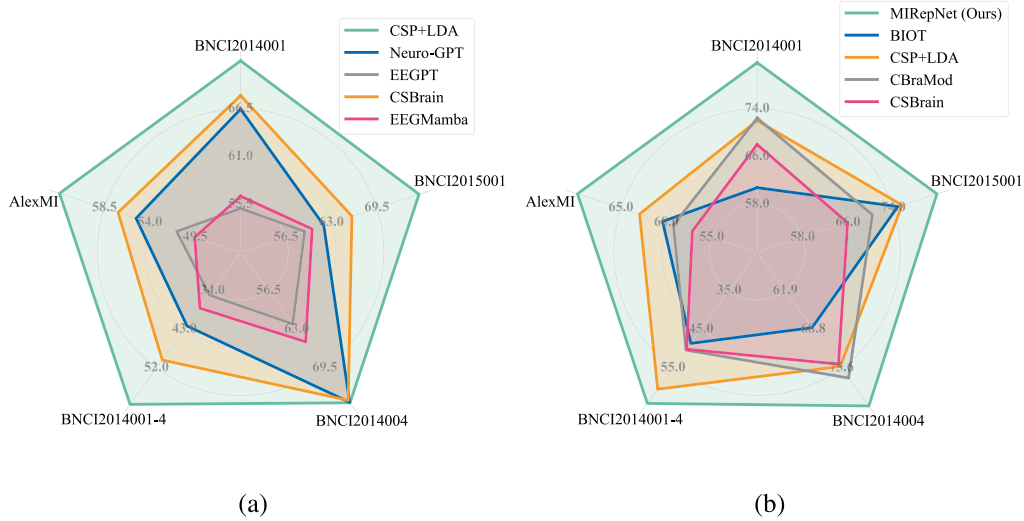


Fig. 2. Classification accuracy (%) on five downstream MI datasets with 30% of each subject's trials used for fine-tuning. Specialist models were trained from scratch on the same data. (a) Comparison of CSP+LDA against recent EEG foundation models; (b) comparison of MIRRepNet (ours) against competitive EEG specialist and foundation model baselines.

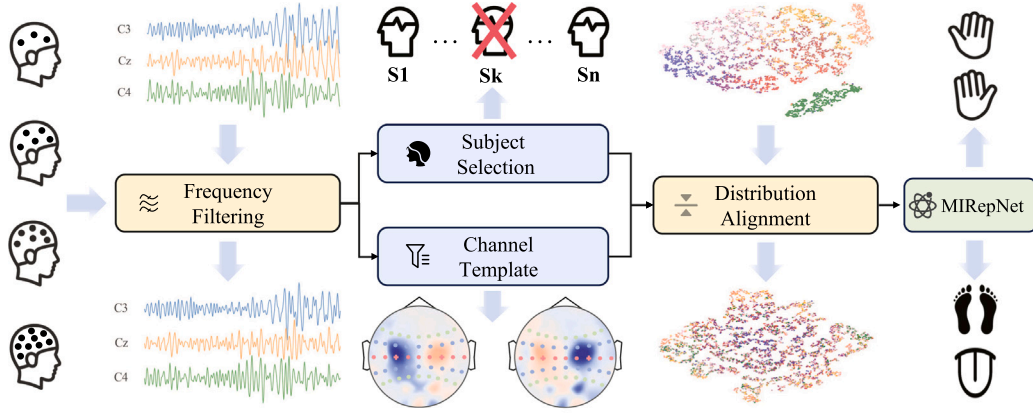


Fig. 3. High-quality MI data pipeline unifying heterogeneous EEG.

LaBraM [11] adopts semantic EEG tokenization and masked modeling to capture generalizable EEG representations. EEGPT [12] leverages spatio-temporal alignment and masked reconstruction tasks to learn universal EEG embeddings, while NeuroLM [27] frames EEG representation learning within a language-modeling paradigm.

However, these mixed-paradigm foundation models often fail to learn truly paradigm-agnostic representations due to fundamental neurophysiological distinctions among different BCI paradigms. Moreover, adapting such generalized models to novel downstream tasks typically necessitates additional post-training involving the entire downstream dataset (including all subjects), significantly increasing data requirements, elevating privacy concerns, and adversely affecting user-friendliness.

3. Method

This section introduces our proposed MIRRepNet, a pre-trained model specifically developed for the MI paradigm. MIRRepNet aims to learn robust and generalizable temporal-spatial EEG representations that facilitate rapid and accurate adaptation to novel MI downstream tasks. To achieve this, we construct a high-quality MI-specific EEG data pipeline (as shown in Fig. 3), introduce a spatially-informed channel template, and propose an efficient pre-training approach combining self-supervised masked reconstruction and supervised MI classification.

3.1. Problem definition

Given r labeled pre-training datasets $\{(X_{i,j}^s, y_{i,j}^s)\}_{i=1}^r$, where $X_{i,j}^s \in \mathbb{R}^{C_s \times T_s^j}$ and $y_{i,j}^s \in \{1, 2, \dots, C\}$ (C is the number of categories), and u downstream target datasets $\{(X_{i,k}^t, y_{i,k}^t)\}_{i=1}^u$, in which $X_{i,k}^t \in \mathbb{R}^{C_t \times T_t^k}$, the goal is to pre-train a model capable of rapidly adaptation to accurately predict the labels $\{y_{i,k}^t\}$ of new trials in downstream MI datasets.

3.2. High-quality MI-EEG data construction

3.2.1. Temporal filtering and alignment

MI decoding primarily leverages modulations in sensorimotor rhythms (SMRs), specifically within the α (8–13 Hz) and β (13–30 Hz) frequency bands [28]. Therefore, we first apply a standard band-pass filter of 8–30 Hz to extract these task-relevant frequency components. Due to varying sampling frequencies across different datasets, we subsequently resample all EEG trials to a unified sampling rate f_{target} , typically set to 250 Hz, for consistent temporal alignment across datasets:

$$\bar{X}_{i,j}^s = \text{Resample}(X_{i,j}^s, f_{\text{orig}}^j, f_{\text{target}}) \in \mathbb{R}^{C_s \times T}, \forall j \in \{1, \dots, r\}. \quad (1)$$

3.2.2. Subject selection

EEG data quality typically exhibits significant variation among subjects due to differences in engagement, fatigue, and/or recording conditions. To maintain high-quality data for model training, we implement a rigorous subject screening protocol. Specifically, for each subject, we train a preliminary within-subject classifier using only their own trials. Subjects whose accuracy falls below a predefined performance threshold are excluded from subsequent analysis. This procedure effectively eliminates data compromised by inattention, artifact contamination, or poor signal integrity, yielding a robust dataset with reliable and stable EEG characteristics.

3.2.3. Channel template

EEG headset configurations differ substantially in terms of electrode number and placement, posing challenges for integrating heterogeneous datasets. Based on MI-specific neurophysiological knowledge, we define a standard channel template consisting of electrodes positioned over the frontal-central (FC), central (C), centro-parietal (CP), and temporal (T) regions, denoted by a fixed set of $\{t_i\}_{i=1}^C$ electrodes. Detailed descriptions of the neurophysiological rationale underlying this design are provided in [Appendix A](#).

For trials from datasets with different electrode configurations, we perform spatial interpolation using inverse-distance weighting. Formally, for each EEG trial $X_{i,j}^s \in \mathbb{R}^{C_s^j \times T}$ recorded from channels $\{e_k\}_{k=1}^{C_s^j}$, we first calculate Euclidean distances between the electrode coordinates:

$$d_{ik} = \|\phi(t_i) - \phi(e_k)\|_2, \quad i = 1, \dots, C, \quad k = 1, \dots, C_s^j, \quad (2)$$

where $\phi(\cdot)$ maps electrode indices to the 2D scalp coordinate.

We then compute the interpolation weights:

$$W_{ik} = \begin{cases} 1, & \exists k^* : d_{ik^*} = 0, \quad k = k^*, \\ 0, & \exists k^* : d_{ik^*} = 0, \quad k \neq k^*, \\ \frac{d_{ik}^{-1}}{\sum_{l=1}^{C_s^j} d_{il}^{-1}}, & \text{otherwise,} \end{cases} \quad (3)$$

where $i = 1, \dots, C$ and $k = 1, \dots, C_s^j$.

Finally, the spatially-aligned EEG trials become:

$$X'[b, i, t] = \sum_{c=1}^{C_s} W_{ij} \tilde{X}[b, c, t], \quad b = 1, \dots, B, \quad i = 1, \dots, C, \quad t = 1, \dots, T. \quad (4)$$

This channel template approach standardizes input EEG signals, preserving MI-related spatial information and enabling unified modeling.

3.2.4. Distribution alignment

EEG signals inherently exhibit non-stationarity and inter-subject variability. To reduce marginal distribution shifts, we adopt Euclidean alignment (EA) [29,30] via whitening transformations. For a given subject with n processed trials $\{X'_i \in \mathbb{R}^{C \times T}\}_{i=1}^n$, we first compute the reference covariance matrix:

$$\bar{R} = \frac{1}{n} \sum_{i=1}^n X'_i (X'_i)^\top, \quad (5)$$

and then normalize each trial by

$$\tilde{X}_i = \bar{R}^{-1/2} X'_i, \quad i = 1, \dots, n. \quad (6)$$

After EA, $\frac{1}{n} \sum_{i=1}^n \tilde{X}_i \tilde{X}_i^\top = I$, so the transformed EEG trials $\{\tilde{X}_i\}$ from different subjects have the identity covariance, effectively mitigating shifts in second-order statistics.

In all subsequent steps, we replace the original trials $\{X_i\}$ with the aligned signals $\{\tilde{X}_i\}$ to ensure a more consistent input distribution. Specifically, in the within-subject few-shot scenario, to prevent data leakage, the EA reference covariance is computed exclusively from the available fine-tuning trials and the resulting reference matrix is applied solely to the fine-tuning data. The test split is aligned independently using its own reference covariance computed from the test trials.

3.3. Temporal-spatial representation

The architecture of MIRepNet is shown in [Fig. 4](#). Given an aligned EEG trial $\tilde{X} \in \mathbb{R}^{C \times T}$, we first encode the temporal dynamics through a convolutional embedding φ_t , with kernel size k_t and stride s_t , yielding:

$$U = \varphi_t(\tilde{X}) = \text{Conv}_t(\tilde{X}) \in \mathbb{R}^{C \times H \times W}, \quad (7)$$

where $W = \lceil T/s_t \rceil$. We then rearrange U to $U' \in \mathbb{R}^{W \times C \times H}$. A subsequent spatial convolutional encoder φ_s compresses the spatial information:

$$S = \varphi_s(U') = \text{Conv}_s(U') \in \mathbb{R}^{W' \times 1 \times H}, \quad (8)$$

where W' depends on the spatial convolution parameters. Finally, we apply temporal average-pooling and a 1×1 convolution to generate compact tokens:

$$Z = \text{Conv}_{1 \times 1}(\text{AvgPool}(S)) \in \mathbb{R}^{D \times 1 \times H'}, \quad (9)$$

reshaped as a sequence of H' tokens $\{z_i \in \mathbb{R}^D\}_{i=1}^{H'}$, each capturing integrated temporal-spatial EEG features.

3.4. MIRepNet pre-training

3.4.1. Masked token reconstruction

After obtaining H' temporal-spatio tokens $\{z_i \in \mathbb{R}^D\}_{i=1}^{H'}$ from each EEG trial, we apply a masked reconstruction task to encourage the model to learn robust contextual representations.

Specifically, let $\alpha \in (0, 1)$ be the mask ratio. We uniformly sample a mask set $\mathcal{M} \subset \{1, \dots, H'\}$ with $|\mathcal{M}| = \alpha H'$, and define the masked token sequence

$$\tilde{z}_i = \begin{cases} [\text{MASK}], & i \in \mathcal{M}, \\ z_i, & i \notin \mathcal{M}, \end{cases} \quad (10)$$

where $[\text{MASK}] \in \mathbb{R}^D$ is a learnable embedding. The masked tokens $\{\tilde{z}_i\}$ are fed into an M -layer Transformer encoder f_θ , producing contextual embeddings $\{c_i\}$. A lightweight Transformer decoder g_ϕ then reconstructs the full token set:

$$\hat{z}_i = g_\phi(\{c_j\}_{j=1}^{H'}), \quad i = 1, \dots, H'. \quad (11)$$

We then minimize the reconstruction loss:

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\hat{z}_i - z_i\|_2^2. \quad (12)$$

By predicting the original tokens from their context, the model learns to capture coherent temporal-spatial patterns, which improves downstream MI decoding performance.

3.4.2. Supervised MI classification

To further enhance the representation discriminability, we attach a supervised classification head $f_{\text{cls}}: \mathbb{R}^D \rightarrow \mathbb{R}^C$ to the Transformer encoder. Given the pooled representation

$$v = \frac{1}{H'} \sum_{i=1}^{H'} c_i \in \mathbb{R}^D, \quad (13)$$

the head produces class logits $s = f_{\text{cls}}(v) \in \mathbb{R}^C$. For a true label $y \in \{1, \dots, C\}$, we define the cross-entropy loss as:

$$\mathcal{L}_{\text{cls}} = -\log \frac{\exp(s_y)}{\sum_{k=1}^C \exp(s_k)}. \quad (14)$$

During pre-training, $\mathcal{L}_{\text{cls}}$ is averaged over all labeled trials and combined with the masked-reconstruction loss to jointly optimize the model.

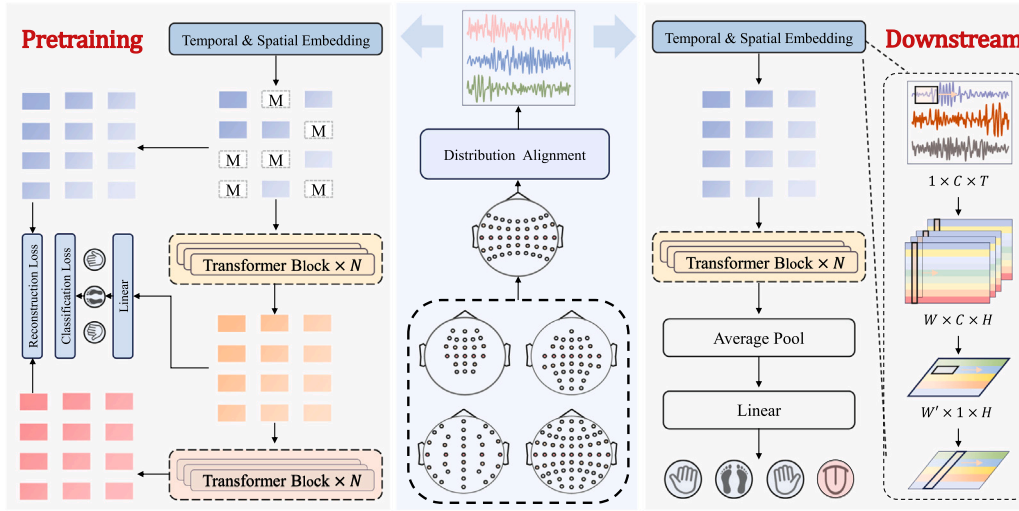


Fig. 4. Overview of MIRepNet. Heterogeneous MI-EEGs from different headsets are first unified to a common channel template and whitened via distribution alignment. The aligned trial is converted into temporal-spatial tokens. A proportion α of tokens are masked and reconstructed with a Transformer encoder-decoder, while the same encoder is simultaneously supervised by an MI classification head. The joint objective learns precise, generalizable MI representations that adapt with only few-shot data.

3.4.3. Joint pre-training

MIRepNet is trained by jointly minimizing the masked token reconstruction loss $\mathcal{L}_{\text{rec}}$ and the MI classification loss $\mathcal{L}_{\text{cls}}$. Formally, the overall pre-training loss is

$$\mathcal{L}_{\text{pretrain}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{cls}}. \quad (15)$$

3.5. Downstream generalization

During downstream generalization, the same preprocessing pipeline described in Section 3.2 is applied to the incoming EEG data, including frequency filtering, channel template mapping, and EA, ensuring distributional consistency with the pre-training stage. For fine-tuning, the Transformer decoder is discarded, and a task-specific classification head is attached to the pre-trained encoder.

4. Experiments

This section performs experiments to validate the effectiveness of MIRepNet.

4.1. Datasets

We conducted experiments utilizing various MI datasets from MOABB [31], summarized in Table 1. Seven publicly available EEG datasets, *BNCI2014002* [32], *PhysionetMI* [33], *Dreyer2023* [34], *Weibo2014* [35], *Zhou2016* [36], *Lee2019* [37], and *Cho2017* [38], were used for pre-training the MIRepNet model. We further evaluated the downstream performance on five independent datasets (47 subjects): *BNCI2014001* [39], *BNCI2015001* [40], *BNCI2014004* [41], *AlexMI* [42], and *BNCI2014001-4* [39] covering diverse electrode configurations, sampling frequencies, and MI tasks.

4.2. Experiment settings

All EEG signals were uniformly resampled to 250 Hz during preprocessing. For MIRepNet pre-training, the masking ratio α was set to 50%. The token embedding dimension D was set to 256, and the Transformer

encoder consisted of 6 layers with a dropout rate of 0.5. We trained MIRepNet for 100 epochs using the Adam optimizer with a learning rate of 1×10^{-3} and a batch size of 64. During downstream adaptation, models were fine-tuned on a limited subset of trials (30% trials in each subject's session) and generally converged within ~ 10 epochs. All reported results were averaged across three runs with different random seeds.

4.3. Main results

We compared MIRepNet with nine specialist algorithms and five existing EEG generalist models (foundation models). Specialist models include CSP+LDA [43], ShallowConv [22], DeepConv [22], EEGNet [21], FBCNet [23], Conformer [25], IFNet [24], ADFCNN [44], and EDPNet [45]. EEG generalized models include BENDR [26], BIOT [9], CBraMod [10], LaBraM [11], Neuro-GPT [15], EEGPT [12], CSBrain [14] and EEGMamba [13]. All EEG foundation models were fine-tuned from their pre-trained weights, whereas the specialist models were trained from scratch using the same fine-tuning data. For the foundation models, we evaluated both full fine-tuning and linear probing strategies.

Table 2 presents the main results, with detailed per-subject results provided in Appendix D. Our proposed MIRepNet consistently achieved the highest average performance across all five downstream datasets, outperforming both specialist and generalized model baselines.

Notably, existing EEG generalist models are pre-trained by mixing multiple BCI paradigms. Because the underlying neurophysiology differs substantially across paradigms, these algorithms struggle to learn truly useful representations and therefore depend on heavy post-training with large amounts of target data. For example, on BNCI2014001, adapting a generalist model to subject i typically requires combining data from the other eight subjects to retrain/fine-tune the network. In contrast, our downstream setting is more stringent (motivation is that the pre-trained model should rapidly adapt to any subject): MIRepNet was fine-tuned with only 30% trials of a single session from the target user (< 30 trials per class) and still achieved precise decoding. In the reported results, generalist baselines were fine-tuned with 80% of the target session.

Table 1
Summary of the pre-training and downstream MI-EEG datasets.

	Dataset	Number of subjects	Number of channels	Sampling rate (Hz)	Number of total trials	Categories
Pre-training	BNCI2014002	14	15	512	2240	right hand, feet
	PhysionetMI	109	64	160	7373	left hand, right hand, feet
	Dreyer2023	87	27	512	16 152	left hand, right hand
	Weibo2014	10	60	200	2370	left hand, right hand, feet
	Zhou2016	4	14	250	900	left hand, right hand, feet
	Lee2019	54	62	1000	10 800	left hand, right hand
	Cho2017	52	64	512	10 520	left hand, right hand
Downstream	BNCI2014001	9	22	250	1296	left hand, right hand
	BNCI2015001	12	13	512	2400	right hand, feet
	BNCI2014004	9	3	250	1400	left hand, right hand
	AlexMI	8	16	512	320	left hand, right hand
	BNCI2014001-4	9	22	250	2592	left hand, right hand, feet, tongue

Table 2
Accuracies (%) in five downstream datasets. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Setting	Approach	# Para	BNCI2014001	BNCI2015001	BNCI2014004	AlexMI	BNCI2014001-4
Full Fine-tuning	Specialist Models (30% Trained)	CSP+LDA	–	72.06	75.54	75.34	62.95	60.45
		ShallowConv	31.6K	68.94 ± 1.15	75.73 ± 0.32	78.21 ± 0.19	56.10 ± 0.42	54.35 ± 0.41
		DeepConv	27.7K	60.07 ± 1.65	61.55 ± 0.34	75.93 ± 1.00	51.19 ± 0.92	31.37 ± 0.10
		EEGNet	1.9K	62.38 ± 2.19	76.09 ± 0.99	73.64 ± 0.21	52.98 ± 0.67	48.63 ± 0.51
		IFNet	9.9K	69.86 ± 0.48	77.32 ± 0.49	78.06 ± 0.84	51.34 ± 1.31	57.74 ± 0.88
		ADFCNN	3.9K	67.66 ± 1.04	76.85 ± 0.84	77.33 ± 0.31	59.38 ± 3.24	52.62 ± 0.45
		Conformer	146.8K	73.67 ± 0.45	77.44 ± 0.64	79.00 ± 0.68	63.54 ± 2.74	59.00 ± 0.69
		FBCNet	1.5K	71.03 ± 1.31	75.81 ± 0.17	76.92 ± 0.13	58.63 ± 1.28	58.54 ± 0.59
		EDPNet	30.6K	64.91 ± 0.86	76.17 ± 0.58	75.94 ± 1.22	50.74 ± 1.52	49.45 ± 0.80
		BENDR	4.0M	57.52 ± 1.34	56.59 ± 0.25	71.70 ± 1.46	53.87 ± 4.64	44.90 ± 1.31
	Generalized Models (30% FT)	BIOT	3.2M	60.82 ± 2.16	74.78 ± 1.41	68.43 ± 0.42	60.42 ± 0.21	48.60 ± 0.72
		CBraMod	4.0M	72.48 ± 0.49	70.30 ± 0.72	77.39 ± 0.35	59.23 ± 0.56	50.34 ± 1.18
		LaBraM	5.8M	60.49 ± 0.57	61.39 ± 0.65	69.76 ± 1.23	54.46 ± 1.59	37.13 ± 0.92
		Neuro-GPT	0.16M	66.49 ± 1.50	61.90 ± 0.90	75.25 ± 1.58	55.36 ± 3.18	42.18 ± 0.23
		EEGPT	25.0M	55.05 ± 1.85	59.17 ± 0.87	62.07 ± 0.74	51.34 ± 1.59	34.99 ± 0.25
		CSBrain	4.9M	68.05 ± 0.81	65.93 ± 0.68	74.92 ± 1.53	57.14 ± 0.36	50.11 ± 0.73
		EEGMamba	3.3M	56.46 ± 0.78	60.24 ± 0.48	65.07 ± 1.86	49.55 ± 1.09	38.04 ± 0.45
	MI-PTM (30% FT)	MIRepNet (Ours)	5.2M	81.77 ± 0.27	81.67 ± 0.26	82.36 ± 0.10	69.79 ± 0.42	64.14 ± 0.31
Linear Probing	Generalized Models (30% FT)	BENDR	4.0M	52.69 ± 1.68	50.95 ± 0.90	55.27 ± 2.40	52.98 ± 3.31	31.43 ± 0.85
		BIOT	3.2M	60.82 ± 1.21	67.80 ± 1.21	67.15 ± 1.64	60.12 ± 1.38	47.95 ± 0.53
		CBraMod	4.0M	63.94 ± 0.09	60.81 ± 0.06	70.38 ± 0.90	58.18 ± 1.80	27.61 ± 0.55
		LaBraM	5.8M	60.31 ± 0.83	59.74 ± 0.86	59.83 ± 1.22	59.23 ± 1.87	35.38 ± 0.45
		Neuro-GPT	0.16M	65.69 ± 1.05	65.60 ± 0.86	73.48 ± 1.76	54.61 ± 4.74	49.82 ± 1.55
		EEGPT	25.0M	55.70 ± 1.96	60.87 ± 2.36	65.88 ± 2.18	52.08 ± 3.72	35.86 ± 0.54
		CSBrain	4.9M	65.76 ± 0.14	65.40 ± 0.36	64.67 ± 0.44	58.63 ± 0.56	48.57 ± 0.38
		EEGMamba	3.3M	54.94 ± 0.85	54.25 ± 0.65	59.02 ± 0.34	50.45 ± 2.22	33.84 ± 0.40
	MI-PTM (30% FT)	MIRepNet (Ours)	5.2M	75.36 ± 2.26	73.35 ± 0.78	76.82 ± 1.41	64.88 ± 4.69	51.94 ± 1.84

4.4. Comparison of different fine-tuning data ratios

An effective pre-trained model should generalize well with only a small amount of calibration data. Nevertheless, existing generalized models tend to under-perform in such low-data regimes.

To disentangle whether this degradation stems primarily from limited fine-tuning samples, we gradually varied the fine-tuning data ratio from 10% to 90% in increments of 20% and evaluated all models under each condition. As illustrated in Fig. 5, the classification accuracy generally improved for most models as more fine-tuning data became available, consistent with our expectation. Despite these gains, achieving reliable performance under minimal calibration or even calibration-free conditions remains a fundamental challenge for the practical deployment of BCI systems, and constitutes an important direction for future research.

4.5. Ablation studies

4.5.1. Analysis of pre-training strategy

To quantify the contribution of our pre-training strategy, we compared MIRepNet with two of its variants:

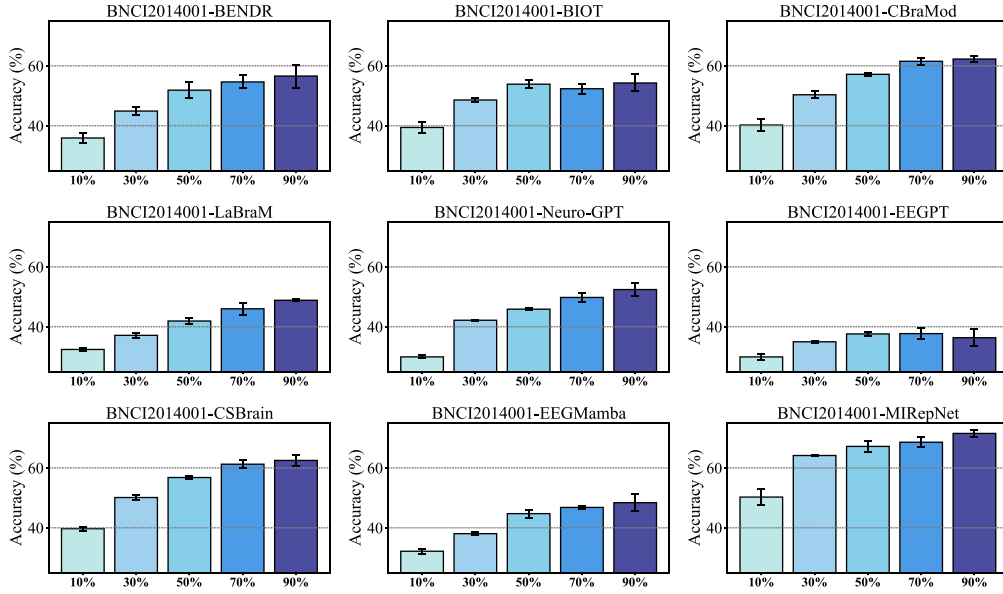
1. w/o Pre-training, trained on 30% trials of the target session.
2. w/o Self-supervised, and the pre-training uses only $\mathcal{L}_{cls}$ (no masked reconstruction).

As shown in Table 3, adding $\mathcal{L}_{rec}$ or the pre-training process always improved the performance.

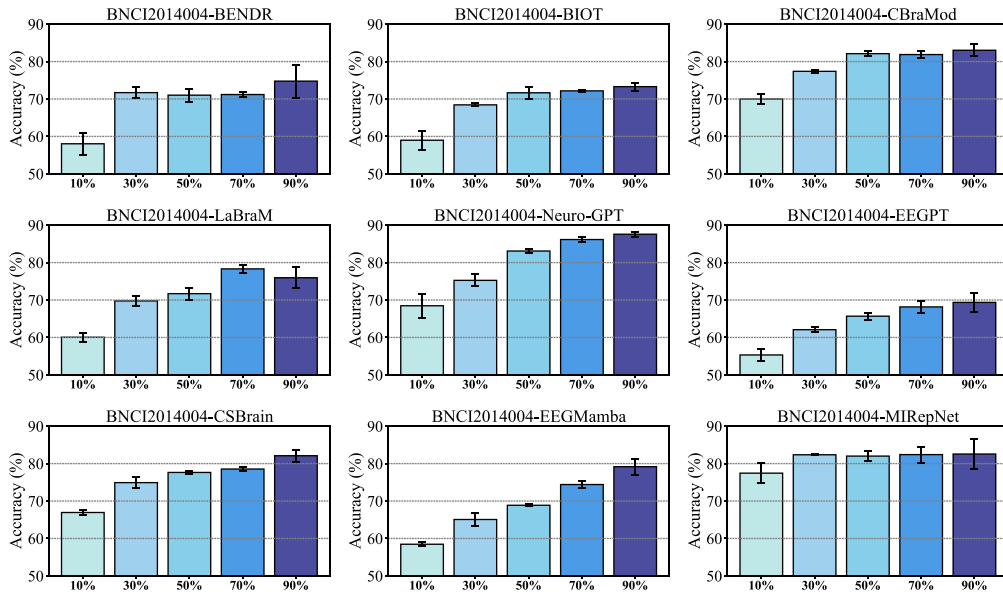
4.5.2. Effectiveness of subject selection and channel template

To evaluate the individual contributions of subject selection and the proposed channel template, we compared MIRepNet with four of its variants:

1. w/o subject selection, pre-trained on all available subjects.
2. w/o channel template-A, pre-trained using only the common electrode subset shared across all pre-training datasets, and downstream data are spatially interpolated to this common subset via inverse-distance weighting.
3. w/o channel template-B, pre-trained using only the common electrode subset, and missing channels in the downstream data are zero-padded to match the common configuration.



(a) Accuracy on BNCI2014001 with different fine-tuning ratios.



(b) Accuracy on BNCI2014004 with different fine-tuning ratios.

Fig. 5. Performance comparison for different fine-tuning data ratios on BNCI2014001 and BNCI2014004 datasets.

4. w/o channel template-C, pre-trained using only the common electrode subset, and an additional convolutional layer is appended during downstream adaptation to project the incoming channels onto the common electrode configuration.

As shown in Table 4, the full MIRRepNet consistently achieved the highest average accuracy on both BNCI2014001-4 and BNCI2014004, demonstrating that both subject selection and the neurophysiologically-informed channel template are essential to the superior performance of MIRRepNet.

4.5.3. Comparison of fine-tuning strategies

To investigate the impact of different fine-tuning strategies on the downstream performance, we compared three protocols:

1. Linear Probing: Only the classification head is updated, while all other parameters remain frozen.
2. Partial Fine-tuning: The convolutional embedding layers and the classification head are updated, while the Transformer encoder remains frozen.
3. Full Fine-tuning: All model parameters are updated.

As shown in Table 5, Full Fine-tuning consistently yielded the best average accuracy on both datasets (64.14% on BNCI2014001-4, and 82.36% on BNCI2014004). Although Linear Probing resulted in reduced performance, it still surpassed existing EEG foundation models reported in Tables D.10 and D.12, demonstrating that the pre-trained representations are inherently discriminative for MI tasks even without encoder adaptation. Partial Fine-tuning further narrowed the gap with

Table 3

Accuracies (%) of ablation studies on five downstream tasks. The best accuracies of each dataset are marked in bold.

Dataset	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	Avg.
BNCI2014001	w/o Pre-training	78.22	62.71	89.77	58.75	75.25	61.39	67.99	91.09	80.53	–	–	–	73.96 _{±1.30}
	w/o Self-supervised	86.47	50.17	94.72	72.28	63.37	63.70	71.95	88.78	77.89	–	–	–	74.37 _{±0.39}
	MIRepNet (All)	92.41	61.39	96.04	77.89	76.24	72.28	79.87	92.08	87.79	–	–	–	81.77 _{±0.27}
BNCI2015001	w/o Pre-training	97.38	91.90	93.81	84.52	85.48	69.05	85.48	57.86	58.10	69.76	88.33	55.00	78.06 _{±0.73}
	w/o Self-supervised	97.86	94.05	92.86	84.76	90.00	74.52	87.62	63.57	65.71	73.33	88.33	55.24	80.65 _{±0.95}
	MIRepNet (All)	97.62	95.00	94.76	85.48	92.86	71.67	86.43	69.52	63.57	76.90	91.43	54.76	81.67 _{±0.26}
BNCI2014004	w/o Pre-training	75.30	48.02	56.55	97.92	85.42	80.06	72.32	83.93	83.04	–	–	–	75.84 _{±1.14}
	w/o Self-supervised	74.11	57.14	53.27	97.62	83.63	82.74	72.62	87.20	87.80	–	–	–	77.35 _{±0.81}
	MIRepNet (All)	85.71	61.51	61.61	98.81	88.99	85.42	79.17	88.39	91.67	–	–	–	82.36 _{±0.10}
AlexMI	w/o Pre-training	47.62	59.52	70.24	79.76	59.52	55.95	91.67	66.67	–	–	–	–	66.37 _{±2.92}
	w/o Self-supervised	52.38	63.10	72.62	63.10	66.67	50.00	92.86	78.57	–	–	–	–	67.41 _{±1.31}
	MIRepNet (All)	55.95	59.52	73.81	70.24	75.00	52.38	96.43	75.00	–	–	–	–	69.79 _{±0.42}
BNCI2014001-4	w/o Pre-training	65.02	51.49	75.58	47.36	33.83	37.62	69.80	78.55	56.11	–	–	–	57.26 _{±0.63}
	w/o Self-supervised	69.14	47.85	82.84	46.53	36.96	43.89	77.72	74.92	56.27	–	–	–	59.57 _{±0.55}
	MIRepNet (All)	71.78	55.12	83.00	52.97	49.34	45.21	78.55	81.02	60.23	–	–	–	64.14 _{±0.31}

Table 4

Accuracies (%) of ablation studies on BNCI2014001-4 and BNCI2014004. The best accuracies of each dataset are marked in bold.

Dataset	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	Avg.
BNCI2014001-4	w/o subject selection	67.16	51.65	76.40	48.51	44.06	41.75	76.24	75.58	59.41	60.08 _{±0.83}
	w/o channel template-A	69.64	55.12	78.22	44.22	37.13	38.94	73.43	70.63	60.73	58.67 _{±1.60}
	w/o channel template-B	68.98	53.96	78.71	42.74	36.63	37.95	72.77	69.80	61.22	58.09 _{±0.93}
	w/o channel template-C	57.76	46.37	62.54	32.01	39.44	35.48	61.55	62.38	48.68	49.58 _{±0.98}
	MIRepNet (All)	71.78	55.12	83.00	52.97	49.34	45.21	78.55	81.02	60.23	64.14 _{±0.31}
BNCI2014004	w/o subject selection	84.23	63.49	59.82	98.21	89.88	83.93	72.62	86.31	88.99	80.83 _{±1.42}
	w/o channel template-A	84.82	65.08	63.69	98.51	86.61	79.46	73.51	89.29	88.99	81.11 _{±0.39}
	w/o channel template-B	83.33	59.52	59.23	97.62	86.90	80.06	76.19	88.69	90.18	80.19 _{±0.98}
	w/o channel template-C	83.33	53.57	55.95	98.21	87.50	81.25	73.81	85.42	88.39	78.60 _{±0.87}
	MIRepNet (All)	85.71	61.51	61.61	98.81	88.99	85.42	79.17	88.39	91.67	82.36 _{±0.10}

Table 5

Accuracies (%) of different fine-tuning strategies on BNCI2014001-4 and BNCI2014004. The best accuracies of each dataset are marked in bold.

Dataset	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	Avg.
BNCI2014001-4	Linear Probing	69.97	47.03	71.78	34.49	37.13	35.31	58.91	59.74	53.14	51.94 _{±1.84}
	Partial Fine-tuning	71.78	54.46	80.53	45.87	45.87	39.44	75.25	77.23	65.51	61.77 _{±0.62}
	Full Fine-tuning	71.78	55.12	83.00	52.97	49.34	45.21	78.55	81.02	60.23	64.14 _{±0.31}
BNCI2014004	Linear Probing	73.21	58.33	63.99	96.43	85.42	70.83	67.26	88.39	87.50	76.82 _{±1.41}
	Partial Fine-tuning	82.44	53.57	57.74	98.21	90.77	84.23	78.87	87.50	88.10	80.16 _{±0.86}
	Full Fine-tuning	85.71	61.51	61.61	98.81	88.99	85.42	79.17	88.39	91.67	82.36 _{±0.10}

Full Fine-tuning, suggesting that the convolutional embedding layers benefit substantially from task-specific adjustment. These results confirmed that the Transformer encoder acquires robust and transferable temporal-spatial representations during pre-training, and that progressively unfreezing more parameters provides consistent performance gains.

4.6. Analysis of the mask ratio

We varied the masking ratio α on downstream datasets (Fig. 6) and observed only minor fluctuations in accuracy, indicating that MIRepNet is robust to this hyperparameter. The performance peaked around $\alpha = 0.5$ and remained competitive for nearby values, confirming that the hybrid objective can effectively exploit both visible and reconstructed tokens.

5. Discussion

This section presents additional discussions.

5.1. Effectiveness of EA

During high-quality data construction, we applied EA to the channel template-unified EEG. From an algebraic standpoint, as illustrated in

Section 3.2.4, EA normalizes each subject's average trial covariance to the identity matrix, thereby eliminating inter-subject discrepancies in second-order statistics. From a Riemannian geometry perspective, the raw covariance matrices of different subjects occupy distinct locations on the symmetric positive-definite manifold. EA re-centers each subject's covariance distribution to the identity matrix, which constitutes the canonical reference point on the symmetric positive-definite manifold, so that when mapped back to Euclidean space, all subjects' trials are mapped into a common distribution space, as visualized in Fig. 7.

5.2. Rapid calibration

A key benefit of MIRepNet is its ability to adapt rapidly to downstream subjects and tasks with minimal data. As shown in Fig. 8, when fine-tuned on only 30% of a new subject's trials, both loss and accuracy converged to near-peak levels within approximately 10 epochs of training. This fast convergence demonstrates MIRepNet's practicality for rapid calibration in motor imagery BCI applications.

5.3. Future research directions

The following directions will be considered in our future research.

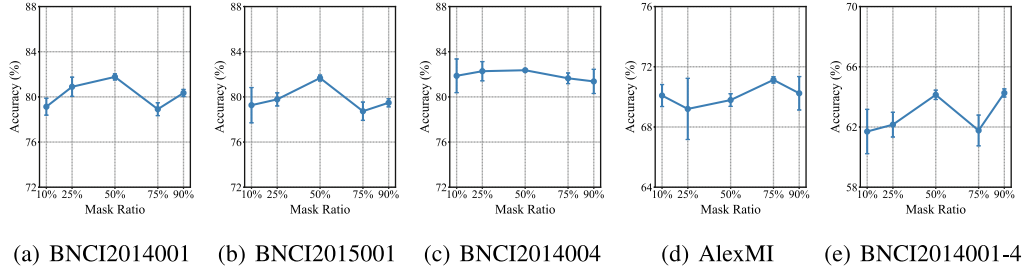


Fig. 6. Sensitivity to mask ratio on the three downstream MI datasets.

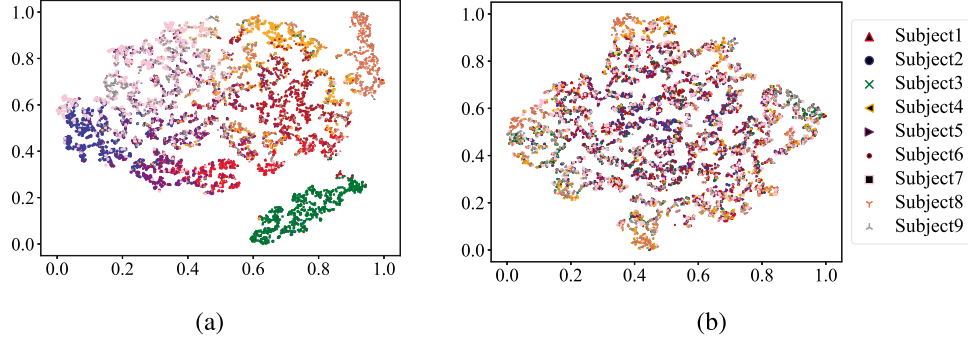


Fig. 7. *t*-SNE visualization of BNCI2014004. (a) Before EA; (b) After EA. Different colors represent trials from different subjects.

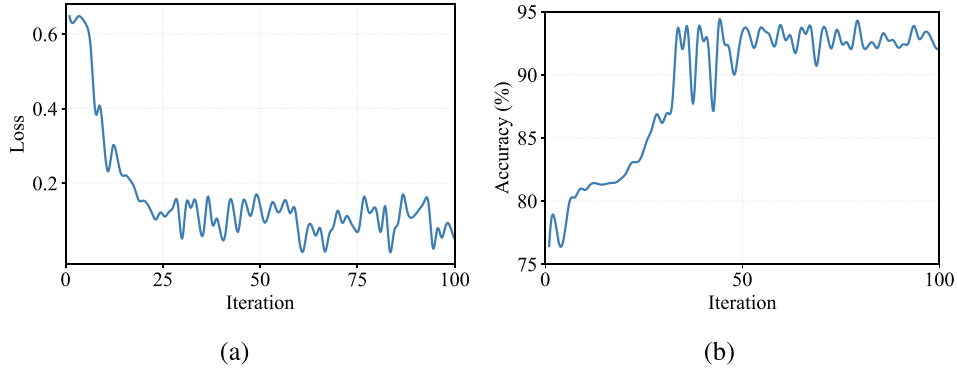


Fig. 8. Test loss and accuracy curves during subject-specific fine-tuning of MIRepNet.

5.3.1. Large-scale high-quality data construction

EEG signals inherently exhibit low signal-to-noise ratios due to hardware limitations and subject variability. Reconstructing raw signals during pre-training, as adopted by many existing methods, may inadvertently encourage models to fit noise rather than learn transferable features. Furthermore, current EEG foundation models have not demonstrated strong generalization ability, potentially due to the scarcity of high-quality data. As verified by Liu et al. [16], training on a curated subset with neurophysiologically-guided quality control can outperform training on larger but uncleaned corpora. Therefore, constructing large-scale yet rigorously curated EEG datasets remains a critical open challenge.

5.3.2. Paradigm-specific foundation models

Since the target paradigm is typically known prior to data acquisition, paradigm-specific foundation models offer a practical alternative to general-purpose ones. Our results in this paper further confirm that restricting pre-training to paradigm-relevant data while incorporating neurophysiological priors yields more discriminative representations for the target task.

5.3.3. Efficient pre-training and fine-tuning strategies

Most existing approaches adopt masked reconstruction as the primary pre-training objective, yet no single strategy has achieved consistently strong generalization across all downstream tasks. Key open problems include handling cross-device heterogeneity more effectively and designing pre-training objectives that yield truly universal representations. Moreover, it is also important to investigate parameter-efficient fine-tuning approaches that enable rapid adaptation to novel tasks, particularly those that maintain strong performance under limited labeled data and reduce the distribution gap between pre-trained representations and target domains.

6. Conclusion

In this paper, we introduce MIRepNet, the first EEG pre-trained model specifically designed for the MI paradigm. A high-quality EEG data pipeline was developed, featuring a neurophysiologically informed channel template that aligns heterogeneous EEG electrode layouts into a unified spatial framework. Furthermore, an efficient pre-training strategy combining self-supervised masked token reconstruction and supervised MI classification was proposed, enabling rapid adaptation to:

Table A.6

Frequency bands and their characteristics in MI paradigm.

Band	Range (Hz)	Associated regions
δ	0.5–4	Deep sleep, unconscious states
θ	4–8	Relaxation, light sleep, meditation, motor imagery
α	8–13	Relaxation, sensorimotor areas, motor imagery
β	13–30	Central sensorimotor regions, motor control
γ	30–45	Higher cognition, sensory processing

(1) new subjects; (2) new tasks; and (3) arbitrary headset configurations (varying in electrode number and placement), achieved via minimal downstream fine-tuning (fewer than 30 trials per class). Extensive evaluations on five downstream MI tasks encompassing 47 subjects demonstrated the efficacy and robustness of MIRepNet, consistently outperforming state-of-the-art specialist and generalist EEG models. Our results underscore the significant advantage and practical necessity of paradigm-specific EEG foundation models.

CRedit authorship contribution statement

Dingkun Liu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation. **Zhu Chen:** Writing – review & editing, Validation, Methodology, Investigation. **Jingwei Luo:** Writing – review & editing, Methodology, Investigation. **Shijie Lian:** Writing – review & editing, Visualization, Investigation. **Yuheng Chen:** Writing – review & editing, Methodology. **Shaojie Hou:** Writing – review & editing, Data curation. **Xiaolian Zhu:** Writing – review & editing, Data curation. **Dongrui Wu:** Writing – review & editing, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Neurological principle of MI

Motor imagery (MI) signals are characterized by two well-established neurophysiological phenomena: event-related desynchronization (ERD) and event-related synchronization (ERS). When a subject imagines performing a movement, the power of specific frequency bands, primarily the α (8–13 Hz) and β (13–30 Hz) bands, decreases in brain regions associated with the imagined action (see Table A.6 for a summary of relevant frequency bands). This power attenuation, termed ERD, is typically observed over the sensorimotor cortex and reflects cortical activation. Conversely, during non-imagery or resting states, certain cortical areas exhibit increased power in these bands, a phenomenon known as ERS. ERD thus serves as a reliable neural correlate of motor planning and intention, whereas ERS is generally associated with cortical idling [46].

The influence of the α and β bands on MI decoding is most prominent over the sensorimotor cortex. During motor imagery, the cortical regions corresponding to the imagined movement exhibit ERD in these bands, whereas regions not engaged in the task tend to display ERS. Importantly, this modulation follows a contralateral organization: left-hand imagery predominantly elicits ERD over the right hemisphere

Table B.7

Architecture hyperparameters of MIRepNet.

	Hyperparameters	Values
Patch Embedding	Input shape	$B \times 45 \times 1000$
	Temporal conv kernel/stride	$(1 \times 25), (1 \times 1)$
	Temporal conv channels (in→out)	$1 \rightarrow 64$
	Spatial conv kernel/stride	$(45 \times 1), (1 \times 1)$
	Spatial conv channels (in→out)	$64 \rightarrow 256$
Tokenization	Pooling kernel/stride	$(1 \times 75), (1 \times 15)$
	Number of tokens L	61
	Token dimension D	256
Transformer Encoder	Number of layers	6
	Number of attention heads	8
	Head dimension d	32
	FFN expansion ratio	4
	FFN hidden dimension	1024
Pre-training Decoder	Mask ratio	0.5
	Number of layers	2
	Number of attention heads	8
Classification Head	Pooling strategy	Mean over tokens
	Output dimension	Task-dependent

(around C4), while right-hand imagery produces ERD over the left hemisphere (around C3). Higher β -band power further indicates more pronounced synchronization in the ipsilateral hemisphere.

Fig. A.9(a) and (b) illustrate this contralateral pattern. In (a), right-hand imagery induces spectral power decreases predominantly over the left sensorimotor cortex; in (b), left-hand imagery produces analogous decreases over the right hemisphere. These lateralized cortical signatures form the neurophysiological basis for BCI systems that distinguish between limb-specific MI tasks.

Guided by these principles, we design a channel template that covers the cortical regions most informative for MI decoding, including the frontal-central (FC), central (C), centro-parietal (CP), and temporal (T) areas. Specifically, the proposed template comprises 45 electrodes: F7, F5, F3, F1, Fz, F2, F4, F6, F8, FT7, FC5, FC3, FC1, FCz, FC2, FC4, FC6, FT8, T7, C5, C3, C1, Cz, C2, C4, C6, T8, TP7, CP5, CP3, CP1, CPz, CP2, CP4, CP6, TP8, P7, P5, P3, P1, Pz, P2, P4, P6, and P8, as illustrated in Fig. A.9(c).

Appendix B. Model architecture

The detailed architectural specifications of MIRepNet are summarized in Table B.7.

Appendix C. Dataset description

The pre-training and downstream datasets used in this work are summarized below:

1. BNCI2014002 includes EEG data from 13 participants performing sustained MI of the right hand and feet. The session consists of eight runs, with 50 trials per class for training and 30 trials for validation. EEG was recorded at 512 Hz from 15 electrodes, including C3, Cz, and C4, with a biosignal amplifier and active Ag/AgCl electrodes.
2. PhysionetMI includes over 1500 one- and two-minute EEG recordings from 109 volunteers performing MI tasks. EEG was recorded with 64 channels using the BCI2000 system.
3. Dreyer2023 dataset concatenates Dreyer2023A/B/C and contains 87 subjects. Each recording includes open/closed-eyes baselines and six MI runs (first two for system acquisition, remaining four for user training), 40 trials per run. Left-/right-hand MI was recorded with 27 electrodes at 512 Hz.

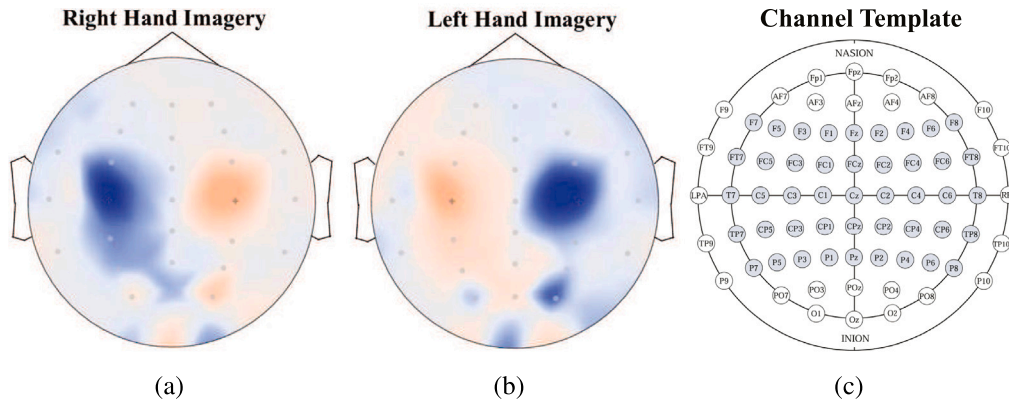


Fig. A.9. Neurophysiological basis and channel template for MI decoding. (a) Scalp topography during right-hand imagery, showing spectral power decreases (blue) predominantly over the left sensorimotor cortex; (b) scalp topography during left-hand imagery, showing power decreases over the right hemisphere; and, (c) the proposed 45-channel template covering frontal-central, central, centro-parietal, and temporal regions.

Table D.8

Accuracies (%) on BNCI2014001. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Setting	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	Avg.
Full Fine-tuning	Specialist Models (30% Trained)	CSP+LDA	81.18	53.47	93.07	63.37	53.47	66.34	66.34	96.04	75.25	72.06
		ShallowConv	75.91	52.81	95.38	54.46	50.17	59.41	57.10	91.75	83.50	68.94 _{±1.15}
		DeepConv	66.34	50.17	80.20	55.78	49.51	54.79	51.16	65.02	67.66	60.07 _{±1.65}
		EEGNet	60.73	52.48	82.51	51.49	55.78	52.81	53.14	68.32	84.16	62.38 _{±2.19}
		IFNet	80.86	51.49	92.08	57.10	55.45	56.11	58.42	90.10	87.13	69.86 _{±0.48}
		ADFCNN	72.94	53.80	91.09	55.45	50.17	56.44	57.76	85.48	85.81	67.66 _{±1.04}
		Conformer	82.84	56.44	95.05	61.06	60.73	62.05	69.31	94.06	81.52	73.67 _{±0.45}
		FBCNet	81.19	53.14	90.43	58.09	54.79	61.06	63.04	91.42	86.14	71.03 _{±1.31}
		EDPNet	70.63	49.51	93.40	55.78	50.83	54.13	53.14	83.83	72.94	64.91 _{±0.86}
	Generalized Models (30% FT)	BENDR	56.21	52.61	52.61	61.11	67.32	54.58	56.86	54.25	62.09	57.52 _{±1.34}
		BIOT	63.73	56.54	66.99	51.31	47.06	56.86	56.54	69.61	78.76	60.82 _{±2.16}
		CBraMod	69.61	56.86	87.91	65.03	68.30	52.94	76.14	88.89	86.60	72.48 _{±0.49}
		LaBraM	57.19	54.25	59.15	59.48	58.82	55.88	62.75	73.53	63.40	60.49 _{±0.57}
		Neuro-GPT	60.13	52.29	80.39	57.52	54.90	65.36	51.63	93.79	82.35	66.49 _{±1.50}
		EEGPT	50.98	53.27	57.19	53.59	48.69	57.19	55.23	58.17	61.11	55.05 _{±1.85}
		CSBrain	59.15	54.58	73.53	66.99	59.48	58.50	58.50	91.83	89.87	68.05 _{±0.81}
		EEGMamba	54.25	43.79	58.17	59.48	57.52	51.96	48.37	74.51	60.13	56.46 _{±0.78}
	Generalized Models (80% FT)	BENDR	52.22	46.67	54.44	67.78	<u>81.11</u>	56.67	76.67	64.44	76.67	64.07 _{±4.66}
		BIOT	78.89	62.22	75.56	68.89	62.22	54.44	54.44	83.33	88.89	69.88 _{±3.37}
		CBraMod	81.11	58.89	<u>95.56</u>	<u>76.67</u>	86.67	70.00	87.78	86.67	92.22	81.73 _{±1.43}
		LaBraM	50.00	53.33	72.22	70.00	<u>81.11</u>	60.00	67.78	77.78	83.33	68.40 _{±0.35}
		Neuro-GPT	<u>84.44</u>	57.78	91.11	67.78	<u>81.11</u>	76.67	<u>84.44</u>	<u>95.56</u>	95.56	81.60 _{±1.77}
		EEGPT	60.00	50.00	61.11	64.44	52.22	52.22	42.22	46.67	76.67	56.17 _{±2.01}
		CSBrain	67.78	52.22	86.67	72.22	<u>81.11</u>	76.67	80.00	85.56	<u>93.33</u>	77.28 _{±2.15}
		EEGMamba	60.00	53.33	80.00	63.33	73.33	66.67	47.78	91.11	<u>67.78</u>	67.04 _{±1.09}
	MI-PTM (30% FT)	MIRepNet (Ours)	92.41	<u>61.39</u>	96.04	77.89	76.24	<u>72.28</u>	79.87	92.08	87.79	81.77 _{±0.27}
Linear Probing	Generalized Models (30% FT)	BENDR	50.98	53.92	52.94	56.21	52.94	54.90	46.73	53.92	51.63	52.69 _{±1.68}
		BIOT	56.86	57.84	67.32	58.82	47.71	50.65	56.54	74.51	77.12	60.82 _{±0.98}
		CBraMod	51.63	54.25	64.71	63.40	57.52	53.59	50.65	90.85	88.89	63.94 _{±0.09}
		LaBraM	57.84	55.23	57.19	59.48	59.48	55.23	59.48	68.63	70.26	60.31 _{±0.83}
		Neuro-GPT	55.56	51.96	81.37	56.86	56.21	68.30	48.37	86.60	85.95	65.69 _{±1.05}
		EEGPT	53.59	57.19	57.52	59.15	50.65	53.27	48.37	61.76	59.80	55.70 _{±1.96}
		CSBrain	61.76	56.54	65.03	62.42	55.23	61.44	61.11	82.68	85.62	65.76 _{±0.14}
		EEGMamba	55.88	43.14	51.31	56.21	62.09	56.21	47.06	63.73	58.82	54.94 _{±0.85}
	Generalized Models (80% FT)	BENDR	58.89	57.78	55.56	56.67	55.56	47.78	54.44	55.56	56.67	55.43 _{±2.19}
		BIOT	65.56	65.56	57.78	63.33	62.22	52.22	67.78	71.11	81.11	65.19 _{±1.98}
		CBraMod	67.78	52.22	72.22	63.33	60.00	60.00	54.44	84.44	<u>95.56</u>	67.78 _{±1.39}
		LaBraM	57.78	55.56	65.56	68.89	<u>72.22</u>	57.78	<u>72.22</u>	71.11	72.22	65.93 _{±1.09}
		Neuro-GPT	<u>78.89</u>	53.33	<u>91.11</u>	77.78	75.56	65.56	73.33	<u>87.78</u>	98.89	78.02 _{±0.97}
		EEGPT	53.33	<u>61.11</u>	55.56	66.67	44.44	52.22	55.56	63.33	75.56	58.64 _{±0.87}
		CSBrain	68.89	54.44	65.56	74.44	61.11	67.78	68.89	77.78	87.78	69.63 _{±1.21}
		EEGMamba	62.22	41.11	57.78	52.22	67.78	62.22	46.67	57.78	70.00	57.53 _{±0.70}
	MI-PTM (30% FT)	MIRepNet (Ours)	87.79	50.50	96.04	<u>74.92</u>	67.66	66.34	70.63	87.13	77.23	<u>75.36</u> _{±2.26}

Table D.9

Accuracies (%) on BNCI2015001. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Setting	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	Avg.
Full Fine-tuning	Specialist Models (30% Trained)	CSP+LDA	93.57	95.00	94.29	82.86	86.43	66.43	80.71	62.14	52.86	59.29	82.14	50.71	75.54
		ShallowConv	94.52	90.71	88.10	73.10	80.71	64.76	82.62	64.52	63.33	60.95	93.57	51.91	75.73 _{±0.32}
		DeepConv	73.10	58.10	63.33	51.91	67.14	48.57	73.33	47.62	56.91	59.52	86.43	52.62	61.55 _{±0.34}
		EEGNet	95.95	94.52	93.10	81.91	81.43	63.57	82.62	55.72	65.00	64.05	82.62	52.62	76.09 _{±0.99}
		IFNet	95.95	95.24	93.57	86.67	78.81	68.10	81.91	57.62	62.86	64.52	88.81	53.81	77.32 _{±0.49}
		ADFCNN	95.71	95.24	90.48	82.62	80.48	67.14	84.05	58.10	58.33	60.48	89.29	60.24	76.85 _{±0.84}
		Conformer	95.71	94.29	88.57	81.67	85.95	68.57	84.52	53.57	63.33	68.33	89.05	55.71	77.44 _{±0.64}
		FBCNet	96.19	94.05	94.05	81.67	82.38	65.24	79.76	51.14	60.71	65.48	88.10	50.24	75.81 _{±0.17}
		EDPNet	96.19	95.24	91.91	89.05	<u>88.57</u>	70.71	77.62	59.29	62.62	56.91	75.48	50.48	76.17 _{±0.58}
	Generalized Models (30% FT)	BENDR	61.67	61.67	54.29	53.81	53.81	55.48	58.81	56.90	52.86	61.43	54.29	54.05	56.59 _{±0.25}
		BIOT	<u>98.57</u>	92.38	92.38	<u>87.14</u>	73.57	55.48	80.72	57.14	52.14	63.57	82.62	62.86	74.78 _{±1.41}
		CBraMod	95.95	90.71	75.24	75.24	79.05	62.14	75.48	54.76	55.48	63.10	61.19	55.24	70.30 _{±0.72}
		LaBraM	84.29	81.19	63.10	61.90	51.43	60.71	64.52	55.24	55.48	50.24	56.43	52.14	61.39 _{±0.65}
		Neuro-GPT	67.62	88.57	67.38	68.33	57.38	48.10	65.71	55.48	51.67	61.19	54.76	56.67	61.90 _{±0.90}
		EEGPT	72.38	78.81	88.10	70.48	50.48	49.76	59.29	54.05	54.52	55.48	55.48	51.19	59.17 _{±0.87}
		CSBrain	89.29	95.71	67.38	66.90	60.71	59.29	70.24	54.29	51.43	60.24	57.86	57.86	65.93 _{±0.68}
		EEGMamba	83.57	75.71	60.71	60.95	51.43	54.05	59.05	54.05	57.14	59.52	54.52	52.14	60.24 _{±0.48}
	Generalized Models (80% FT)	BENDR	70.00	68.33	61.67	49.17	62.50	65.00	68.33	60.83	57.50	72.50	65.00	65.83	63.89 _{±1.36}
		BIOT	100.00	<u>97.50</u>	<u>96.67</u>	85.83	65.83	70.83	80.83	61.67	<u>64.17</u>	64.17	85.83	57.50	77.57 _{±1.58}
		CBraMod	97.50	97.50	93.33	82.50	80.00	77.50	75.83	56.67	61.67	68.33	80.83	65.00	<u>78.06</u> _{±1.47}
		LaBraM	97.50	94.17	65.83	74.17	60.83	63.33	64.17	<u>69.17</u>	60.00	65.83	51.67	62.50	69.10 _{±0.77}
		Neuro-GPT	63.33	<u>95.83</u>	99.17	85.83	84.17	<u>76.67</u>	86.67	57.50	57.50	<u>73.33</u>	76.67	<u>67.50</u>	77.01 _{±1.82}
		EEGPT	78.33	92.50	65.00	74.17	51.67	57.50	57.50	49.17	55.83	62.50	62.50	53.33	63.33 _{±2.07}
		CSBrain	98.33	97.50	82.50	72.50	74.17	70.83	74.17	62.50	57.50	67.50	74.17	75.00	75.56 _{±2.25}
		EEGMamba	92.50	94.17	73.33	78.33	61.67	64.17	71.67	61.67	54.17	64.17	67.50	75.00	71.53 _{±2.10}
	MI-PTM (30% FT)	MIRepNet (Ours)	97.62	95.00	94.76	85.48	92.86	71.67	<u>86.43</u>	69.52	63.57	76.90	<u>91.43</u>	54.76	81.67 _{±0.26}
Linear Probing	Generalized Models (30% FT)	BENDR	52.38	50.00	54.76	47.86	54.05	45.00	49.29	50.71	47.62	52.38	54.52	52.86	50.95 _{±0.90}
		BIOT	74.05	95.00	88.81	80.47	63.57	62.62	<u>78.10</u>	49.29	58.10	53.09	59.29	51.19	67.80 _{±1.21}
		CBraMod	91.67	57.86	57.86	60.24	52.38	<u>72.14</u>	60.48	53.57	53.10	49.76	65.24	55.48	60.81 _{±0.06}
		LaBraM	77.62	71.91	63.57	62.86	50.48	61.91	59.28	57.38	54.76	50.00	53.81	53.33	59.74 _{±0.86}
		Neuro-GPT	50.24	91.67	89.52	74.05	63.81	48.10	79.52	57.14	53.57	66.19	60.95	52.38	65.60 _{±0.86}
		EEGPT	78.81	79.52	69.29	67.86	53.57	57.86	63.57	47.62	53.81	56.90	53.10	48.57	60.87 _{±2.36}
		CSBrain	94.52	93.57	67.38	65.95	56.43	63.33	65.24	54.52	54.05	62.38	50.95	56.43	65.40 _{±0.36}
		EEGMamba	66.67	60.00	58.33	54.29	49.76	53.10	53.10	50.48	51.19	47.14	56.67	50.24	54.25 _{±0.65}
	Generalized Models(80% FT)	BENDR	55.83	47.50	61.67	44.17	60.00	42.50	57.50	51.67	53.33	58.33	57.50	55.00	53.75 _{±1.80}
		BIOT	79.17	<u>96.67</u>	96.67	<u>87.50</u>	67.50	66.67	61.67	64.17	58.33	56.67	72.50	51.67	71.60 _{±2.87}
		CBraMod	92.50	57.50	54.17	<u>76.67</u>	57.50	65.00	55.83	54.17	47.50	60.83	65.83	55.83	61.94 _{±0.26}
		LaBraM	90.83	82.50	62.50	73.33	68.33	62.50	67.50	50.83	57.50	50.00	60.83	59.17	65.49 _{±0.35}
		Neuro-GPT	66.67	97.50	91.67	83.33	65.83	67.50	77.50	<u>63.33</u>	51.67	68.33	<u>71.67</u>	61.67	<u>72.22</u> _{±0.55}
		EEGPT	86.67	90.00	71.67	81.67	57.50	55.83	59.17	55.83	65.00	55.00	70.00	48.33	66.39 _{±0.71}
		CSBrain	99.17	95.83	81.67	88.33	55.00	80.83	59.17	53.33	56.67	60.83	72.50	56.67	71.67 _{±0.17}
		EEGMamba	67.50	75.83	62.50	63.33	52.50	54.17	63.33	48.33	55.00	60.83	58.33	61.67	60.28 _{±0.94}
	MI-PTM (30% FT)	MIRepNet (Ours)	<u>96.19</u>	89.76	<u>93.10</u>	81.67	85.24	67.62	70.24	54.29	<u>61.19</u>	<u>66.90</u>	65.48	48.57	73.35 _{±0.78}

- Weibo2014 includes EEG data from 10 subjects recorded with 60 electrodes. It consists of seven mental tasks, including simple and compound limb MI tasks (left hand, right hand, feet, and combinations), and a rest state.
- Zhou2016 includes EEG data from 4 subjects performing three MI tasks: left hand, right hand, and feet. Each subject participated in three sessions, with each session consisting of two runs of 75 trials (25 trials per class).
- Lee2019 includes EEG data recorded from 62 channels at 1000 Hz using a BrainAmp amplifier, which involved MI tasks for left and right hand grasping, with 100 trials per session. The EEG channels were referenced to the nasion and grounded to AFz.
- Cho2017 includes EEG data from 52 subjects (19 females, mean age 24.8 ± 3.86 years) performing MI tasks for the left and right hands. EEG was recorded at 512 Hz from 64 channels using the Biosemi ActiveTwo system, with a 10–10 system montage.
- BNCI2014001 contains EEG data from 9 subjects performing four MI tasks: left hand, right hand, both feet, and tongue. Each subject participated in two sessions, with each session consisting of 6 runs, yielding a total of 288 trials per session.
- BNCI2015001 contains EEG data from subjects performing sustained MI of the right hand and both feet. The data were recorded at 512 Hz using 15 electrodes, including C3, Cz, and

C4, with a bandpass filter between 0.5 and 100 Hz and a notch filter at 50 Hz.

- BNCI2014004 includes EEG data from 9 right-handed subjects, who performed two MI tasks: left hand and right hand. Each subject participated in five sessions, with the first two for screening without feedback and the last three with feedback. The data was recorded with three bipolar EEG channels (C3, Cz, C4) at 250 Hz and included 120 trials per subject for each MI category.
- AlexMI contains EEG recordings from 8 subjects, performing 2 task of motor imagination (right hand, feet or rest). Data have been recorded at 512 Hz with 16 wet electrodes.

Appendix D. Detailed results

The per-subject classification results on all downstream datasets are reported in [Tables D.8–D.12](#).

Appendix E. Efficiency analysis

To assess the computational efficiency of MIRepNet relative to existing EEG foundation models, we measured the forward-pass floating-point operations (FLOPs) for a single trial using a unified input configuration.

Table D.10

Accuracies (%) on BNCI2014004. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Setting	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	Avg.
Full Fine-tuning	Specialist Models (30% Trained)	CSP+LDA	82.14	50.40	53.87	97.32	82.74	79.76	63.69	85.71	82.44	75.34
		ShallowConv	<u>83.04</u>	59.52	51.49	<u>98.51</u>	90.48	71.73	78.87	84.52	85.71	78.21 ± 0.19
		DeepConv	77.08	53.57	52.68	<u>98.51</u>	87.20	67.26	74.70	84.23	88.10	75.93 ± 1.00
		EEGNet	74.70	55.95	52.68	97.92	83.04	68.75	66.97	84.52	78.27	73.64 ± 0.21
		IFNet	81.55	51.98	51.19	97.62	88.10	79.76	76.49	<u>86.31</u>	<u>89.58</u>	78.06 ± 0.84
		ADFCNN	75.89	57.54	55.06	98.21	89.88	71.43	75.89	85.42	86.61	77.33 ± 0.31
		Conformer	81.25	57.14	51.49	98.21	92.56	82.74	78.27	83.33	86.01	79.00 ± 0.68
		FBCNet	79.46	50.00	51.49	<u>98.51</u>	87.20	81.55	73.21	86.01	84.82	76.92 ± 0.13
		EDPNet	68.45	55.16	53.57	98.21	85.71	81.25	75.30	83.04	82.74	75.94 ± 1.22
	Generalized Models (30% FT)	BENDR	65.18	56.35	<u>79.76</u>	53.87	100.00	83.63	91.67	63.69	51.19	71.70 ± 1.46
		BIOT	69.35	49.21	51.49	90.48	88.10	63.69	59.82	73.81	69.94	68.43 ± 0.42
		CBraMod	66.37	50.40	54.17	98.21	98.81	82.14	78.27	82.74	85.42	77.39 ± 0.35
		LaBraM	61.61	48.41	58.04	96.43	97.02	50.30	74.11	83.93	58.04	69.76 ± 1.23
		Neuro-GPT	66.07	57.94	61.61	78.57	98.81	76.49	79.46	86.01	72.32	75.25 ± 1.58
		EEGPT	58.63	55.95	55.36	57.14	83.04	61.01	63.69	62.20	61.61	62.07 ± 0.74
		CSBrain	69.35	61.51	63.10	70.24	98.81	81.85	88.10	75.60	65.77	74.92 ± 1.53
		EEGMamba	56.55	56.75	59.82	62.50	92.26	65.77	74.70	69.64	47.62	65.07 ± 1.86
	Generalized Models (80% FT)	BENDR	61.46	76.39	71.88	57.29	100.00	<u>90.62</u>	94.79	70.83	62.50	76.20 ± 3.18
		BIOT	64.58	48.61	53.12	87.50	100.00	72.92	71.88	77.08	84.38	73.34 ± 1.89
		CBraMod	69.79	<u>75.00</u>	80.21	77.08	100.00	85.42	91.67	71.88	83.33	<u>81.60</u> ± 1.45
		LaBraM	66.67	<u>63.89</u>	67.71	71.88	100.00	83.33	<u>93.75</u>	63.54	80.21	76.77 ± 1.66
		Neuro-GPT	54.17	66.67	71.88	89.58	<u>98.96</u>	92.71	91.67	80.21	85.42	81.25 ± 1.00
		EEGPT	68.75	52.78	66.67	61.46	91.67	71.88	89.58	57.29	55.21	68.36 ± 0.76
		CSBrain	73.96	62.50	68.75	73.96	100.00	83.33	<u>93.75</u>	83.33	80.21	79.98 ± 1.50
		EEGMamba	66.67	<u>75.00</u>	65.62	63.54	<u>98.96</u>	84.38	87.50	65.62	69.79	75.23 ± 1.48
	MI-PTM (30% FT)	MIRepNet (Ours)	85.71	61.51	61.61	98.81	88.99	85.42	79.17	88.39	91.67	82.36 ± 0.10
Linear Probing	Generalized Models (30% FT)	BENDR	52.68	51.59	51.79	50.00	68.75	53.27	61.90	58.04	49.40	55.27 ± 2.40
		BIOT	68.15	50.79	52.08	83.63	82.74	61.61	52.08	77.68	75.60	67.15 ± 1.64
		CBraMod	56.55	52.78	53.27	<u>89.88</u>	95.54	65.18	56.85	<u>83.93</u>	79.46	70.38 ± 0.90
		LaBraM	56.85	49.21	56.25	<u>77.08</u>	65.48	47.92	59.82	73.51	52.38	59.83 ± 1.22
		Neuro-GPT	64.58	61.90	61.31	70.24	98.21	<u>80.95</u>	79.17	82.74	62.20	73.48 ± 1.76
		EEGPT	59.23	61.11	58.93	68.45	88.39	65.18	61.61	67.56	62.50	65.88 ± 1.18
		CSBrain	60.71	<u>62.70</u>	55.95	55.06	98.51	56.55	65.18	71.43	55.95	64.67 ± 0.44
		EEGMamba	50.60	57.94	51.79	57.74	83.33	60.12	58.63	66.96	44.05	59.02 ± 0.34
	Generalized Models (80% FT)	BENDR	60.42	51.39	58.33	52.08	96.88	64.58	67.71	56.25	52.08	62.19 ± 1.29
		BIOT	66.67	43.06	52.08	77.08	91.67	69.79	61.46	82.29	79.17	69.25 ± 0.88
		CBraMod	56.25	54.17	58.33	63.54	<u>98.96</u>	60.42	68.75	68.75	68.75	66.44 ± 3.37
		LaBraM	61.46	41.67	<u>62.50</u>	67.71	93.75	67.71	<u>81.25</u>	63.54	63.54	67.01 ± 0.41
		Neuro-GPT	57.29	66.67	61.46	63.54	100.00	89.58	91.67	76.04	86.46	76.97 ± 1.32
		EEGPT	75.00	55.56	58.33	68.75	97.92	65.62	79.17	70.83	62.50	70.41 ± 1.62
		CSBrain	66.67	58.33	57.29	46.88	100.00	66.67	76.04	64.58	55.21	65.74 ± 0.81
		EEGMamba	64.58	58.33	34.38	54.17	92.71	64.58	59.38	61.46	57.29	60.76 ± 0.71
	MI-PTM (30% FT)	MIRepNet (Ours)	<u>73.21</u>	58.33	63.99	96.43	85.42	70.83	67.26	88.39	87.50	<u>76.82</u> ± 1.41

Table D.11

Accuracies (%) on AlexMI. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Setting	Approach	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
Full Fine-tuning	Specialist Models (30% Trained)	CSP+LDA	42.86	60.71	60.71	64.29	67.86	60.71	92.86	53.57	62.95
		ShallowConv	38.10	46.43	60.71	55.95	50.00	51.19	84.52	61.91	56.10 _{±0.42}
		DeepConv	53.57	50.00	50.00	61.91	52.38	39.29	45.24	57.14	51.19 _{±0.92}
		EEGNet	41.67	53.57	53.57	67.86	46.43	42.86	69.05	48.81	52.98 _{±5.67}
		IFNet	39.29	50.00	51.19	53.57	48.81	47.62	48.81	71.43	51.34 _{±1.31}
		ADFCNN	47.62	44.05	61.91	55.95	53.57	53.57	86.91	71.43	59.38 _{±3.24}
		Conformer	41.67	66.67	72.62	59.52	63.10	46.43	85.71	72.62	63.54 _{±2.74}
		FBCNet	40.48	<u>64.29</u>	70.24	66.67	57.14	40.48	61.91	67.86	58.63 _{±1.28}
		EDPNet	39.29	61.91	60.71	60.71	46.43	42.86	51.19	42.86	50.74 _{±1.52}
	Generalized Models (30% FT)	BENDR	50.00	46.43	53.57	53.57	61.90	55.95	53.57	55.95	53.87 _{±4.64}
		BIOT	57.14	53.57	50.00	53.57	65.48	55.95	75.00	72.62	60.42 _{±0.21}
		CBraMod	51.19	41.67	45.24	54.76	55.95	48.81	94.05	<u>82.14</u>	59.23 _{±0.56}
		LaBraM	57.14	44.05	59.52	54.76	54.76	53.57	64.29	<u>47.62</u>	54.46 _{±1.59}
		Neuro-GPT	48.81	39.29	52.38	53.57	54.76	58.33	78.57	57.14	55.36 _{±3.18}
		EEGPT	48.81	44.05	45.24	52.38	54.76	51.19	59.52	54.76	51.34 _{±1.59}
		CSBrain	51.19	50.00	48.81	61.90	53.57	53.57	66.67	71.43	57.14 _{±0.36}
		EEGMamba	45.24	33.33	46.43	55.95	54.76	52.38	59.52	48.81	49.55 _{±1.09}
	Generalized Models (80% FT)	BENDR	75.00	62.50	37.50	50.00	58.33	<u>62.50</u>	54.17	70.83	58.85 _{±3.21}
		BIOT	<u>70.83</u>	58.33	75.00	33.33	66.67	79.17	83.33	83.33	68.75 _{±3.38}
		CBraMod	54.17	33.33	62.50	91.67	66.67	37.50	83.33	83.33	64.06 _{±5.85}
		LaBraM	58.33	41.67	62.50	62.50	62.50	50.00	70.83	66.67	59.38 _{±5.56}
		Neuro-GPT	50.00	37.50	45.83	54.17	66.67	58.33	<u>95.83</u>	70.83	59.90 _{±2.66}
		EEGPT	58.33	45.83	62.50	41.67	50.00	50.00	54.17	58.33	52.60 _{±3.90}
		CSBrain	45.83	50.00	45.83	58.33	87.50	<u>62.50</u>	91.67	66.67	63.54 _{±5.16}
		EEGMamba	<u>70.83</u>	41.67	41.67	41.67	66.67	58.33	79.17	58.33	57.29 _{±0.74}
	MI-PTM (30% FT)	MIRepNet (Ours)	55.95	59.52	<u>73.81</u>	<u>70.24</u>	<u>75.00</u>	52.38	96.43	75.00	69.79 _{±0.42}
Linear Probing	Generalized Models (30% FT)	BENDR	50.00	53.57	55.95	59.52	46.43	52.38	51.19	54.76	52.98 _{±3.31}
		BIOT	52.38	53.57	51.19	54.76	67.86	47.62	80.95	72.62	60.12 _{±1.38}
		CBraMod	54.76	46.43	46.43	47.62	59.52	53.57	77.38	79.76	58.18 _{±1.80}
		LaBraM	53.57	58.33	<u>66.67</u>	59.52	52.38	<u>58.33</u>	66.67	58.33	59.23 _{±1.87}
		Neuro-GPT	45.24	41.67	53.57	54.76	58.33	53.57	73.81	55.95	54.61 _{±4.74}
		EEGPT	52.38	51.19	51.19	48.81	47.62	52.38	60.71	52.38	52.08 _{±3.72}
		CSBrain	50.00	52.38	46.43	57.14	59.52	57.14	71.43	75.00	58.63 _{±0.56}
		EEGMamba	53.57	42.86	50.00	51.19	46.43	51.19	57.14	51.19	50.45 _{±2.22}
	Generalized Models (80% FT)	BENDR	54.17	41.67	54.17	54.17	62.50	<u>58.33</u>	54.17	54.17	54.17 _{±0.74}
		BIOT	58.33	50.00	<u>66.67</u>	45.83	<u>70.83</u>	62.50	58.33	75.00	60.94 _{±1.28}
		CBraMod	45.83	41.67	62.50	45.83	54.17	54.17	75.00	83.33	57.81 _{±0.00}
		LaBraM	<u>66.67</u>	66.67	70.83	70.83	50.00	62.50	58.33	70.83	64.58 _{±2.66}
		Neuro-GPT	50.00	50.00	70.83	50.00	50.00	41.67	75.00	75.00	57.81 _{±5.56}
		EEGPT	50.00	58.33	54.17	54.17	58.33	45.83	50.00	45.83	52.08 _{±0.74}
		CSBrain	54.17	37.50	70.83	50.00	58.33	54.17	70.83	75.00	58.85 _{±2.95}
		EEGMamba	75.00	50.00	37.50	16.67	<u>70.83</u>	62.50	<u>79.17</u>	66.67	57.29 _{±0.74}
	MI-PTM (30% FT)	MIRepNet (Ours)	59.52	<u>59.52</u>	58.33	<u>60.71</u>	80.95	42.86	76.19	<u>80.95</u>	64.88 _{±4.69}

Table D.12

Accuracies (%) on BNCI2014001-4. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Setting	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	Avg.
Full Fine-tuning	Specialist Models (30% Trained)	CSP+LDA	72.28	55.94	<u>77.72</u>	40.10	35.15	44.06	<u>78.22</u>	76.24	64.36	60.45
		ShallowConv	68.65	51.82	67.33	39.60	32.67	39.77	56.93	65.68	66.67	54.35 _{±0.41}
		DeepConv	31.52	27.23	43.89	28.38	24.59	27.39	23.74	26.57	47.03	31.37 _{±0.10}
		EEGNet	59.41	35.81	71.45	32.01	33.33	32.84	58.09	52.31	62.38	48.63 _{±0.51}
		IFNet	70.63	56.27	71.78	38.61	34.98	37.95	68.48	69.97	70.96	57.74 _{±0.88}
		ADFCNN	64.36	50.17	67.00	39.11	34.16	32.51	63.70	59.24	63.37	52.62 _{±0.45}
		Conformer	69.80	48.68	72.77	44.39	37.46	45.88	71.78	79.04	61.22	59.00 _{±0.69}
		FBCNet	72.28	54.95	77.39	41.09	34.32	39.27	70.46	<u>72.61</u>	64.52	58.54 _{±0.59}
		EDPNet	64.03	37.79	69.14	36.47	30.53	32.84	48.68	67.66	57.92	49.45 _{±0.80}
	Generalized Models (30% FT)	BENDR	36.76	38.89	48.69	38.40	55.88	42.65	46.57	45.75	50.49	44.90 _{±1.31}
		BIOT	61.93	43.79	56.54	30.07	36.76	34.31	59.97	59.31	54.74	48.60 _{±0.72}
		CBraMod	56.37	46.57	69.61	38.40	39.38	30.07	56.54	61.93	54.25	50.34 _{±1.18}
		LaBraM	43.46	32.03	37.75	30.07	34.15	30.39	35.13	43.95	47.22	37.13 _{±0.92}
		Neuro-GPT	52.12	29.08	53.59	36.27	28.92	35.29	36.27	59.31	48.69	42.18 _{±0.23}
		EEGPT	46.08	28.27	33.66	31.05	26.96	32.84	31.37	41.01	43.63	34.99 _{±0.25}
		CSBrain	52.12	45.26	64.54	37.09	42.16	30.88	51.31	65.20	62.42	50.11 _{±0.73}
		EEGMamba	45.42	33.82	44.12	37.09	31.86	32.35	29.74	45.92	41.99	38.04 _{±0.45}
	Generalized Models (80% FT)	BENDR	46.11	32.78	46.11	45.56	53.89	45.00	48.89	43.33	57.78	46.60 _{±1.29}
		BIOT	61.11	47.78	56.67	35.56	<u>61.67</u>	39.44	67.22	61.11	60.56	54.57 _{±2.00}
		CBraMod	67.78	50.56	73.89	52.22	64.44	<u>53.89</u>	71.67	70.00	71.11	63.95 _{±1.30}
		LaBraM	56.11	34.44	58.33	41.11	47.78	37.22	43.33	57.22	55.56	47.90 _{±1.40}
		Neuro-GPT	63.33	46.11	75.00	59.44	57.78	50.56	62.22	76.11	75.56	62.90 _{±1.45}
		EEGPT	44.44	30.56	42.78	36.11	40.00	28.33	34.44	35.56	52.78	38.33 _{±1.18}
		CSBrain	65.56	<u>56.11</u>	76.67	51.67	56.11	54.44	66.11	67.78	<u>72.22</u>	62.96 _{±1.29}
		EEGMamba	50.00	31.11	56.67	42.22	57.22	32.78	51.67	58.33	51.67	47.96 _{±0.40}
	MI-PTM (30% FT)	MIRepNet (Ours)	<u>71.78</u>	55.12	83.00	<u>52.97</u>	49.34	45.21	78.55	81.02	60.23	64.14 _{±0.31}
Linear Probing	Generalized Models (30% FT)	BENDR	31.86	30.72	33.66	33.33	30.07	26.31	34.48	27.61	34.80	31.43 _{±0.85}
		BIOT	56.21	41.34	53.43	37.25	41.01	38.56	52.94	60.46	50.33	47.95 _{±0.53}
		CBraMod	28.59	28.10	27.29	29.74	26.80	26.47	25.65	28.10	27.78	27.61 _{±0.55}
		LaBraM	39.54	29.25	39.38	28.76	36.44	29.58	35.46	37.75	42.32	35.38 _{±0.45}
		Neuro-GPT	55.23	41.01	60.29	42.81	36.11	41.18	46.08	71.24	54.41	49.82 _{±1.55}
		EEGPT	46.08	26.80	35.29	32.68	28.27	29.25	33.99	44.93	45.42	35.86 _{±0.54}
		CSBrain	50.98	<u>44.12</u>	52.29	38.24	37.09	41.83	49.35	62.09	61.11	48.57 _{±0.38}
		EEGMamba	35.95	29.08	42.32	31.05	28.92	30.56	29.25	37.58	39.87	33.84 _{±0.40}
	Generalized Models (80% FT)	BENDR	31.11	30.56	28.33	33.33	37.22	28.33	37.78	26.67	41.67	32.78 _{±0.55}
		BIOT	51.67	39.44	56.11	41.11	<u>47.22</u>	36.11	54.44	57.78	53.89	48.64 _{±0.92}
		CBraMod	37.78	39.44	53.89	<u>44.44</u>	39.44	40.56	32.78	<u>70.00</u>	72.78	47.90 _{±0.75}
		LaBraM	51.67	28.33	51.11	40.00	46.67	45.00	53.33	52.78	57.22	47.35 _{±0.75}
		Neuro-GPT	<u>60.00</u>	33.33	<u>65.00</u>	45.56	41.11	41.11	50.00	68.89	60.56	<u>51.73</u> _{±0.46}
		EEGPT	48.89	27.22	35.56	37.78	35.00	32.22	38.89	47.78	55.00	39.81 _{±1.06}
		CSBrain	55.00	38.33	48.33	41.11	38.89	47.78	<u>55.56</u>	67.22	<u>65.56</u>	50.86 _{±0.98}
		EEGMamba	39.44	30.00	45.00	30.56	41.67	43.33	35.56	38.33	55.56	39.94 _{±0.57}
	MI-PTM (30% FT)	MIRepNet (Ours)	69.97	47.03	71.78	34.49	37.13	35.31	58.91	59.74	53.14	51.94 _{±1.84}

Data availability

All datasets used in this article are publicly available datasets, and download links are provided.

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Table E.13

Computational cost (forward-pass GFLOPs) and parameter counts of MIRepNet and baseline EEG foundation models.

Model	#Params	GFLOPs
Neuro-GPT	0.16M	0.13
BENDR	4.0M	0.53
BIOT	3.2M	1.05
CBraMod	4.0M	1.09
LaBraM	5.8M	1.40
CSBrain	4.9M	1.90
EEGPT	25.0M	20.97
MIRepNet (Ours)	5.2M	1.46

As shown in Table E.13, MIRepNet required approximately 1.46 GFLOPs per forward pass, comparable to models of similar scale such as BIOT (1.05 GFLOPs) and CBraMod (1.09 GFLOPs), and substantially lower than larger models such as EEGPT (20.97 GFLOPs). Despite having a moderate computational cost, MIRepNet consistently achieved superior downstream performance across all five evaluation datasets, demonstrating a favorable trade-off between computational efficiency and classification accuracy.

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